

An Integrated Blockchain-Based Certificate Issuance and Verification Platform Using Ethereum Smart Contracts and Automated PDF Delivery

Ayush Joshi^{*1}, Priyanshu[†], Rohan[†], Neha Gupta^{*2}

^{*1}Students, School of Engineering (SET), Sushant University, Gurugram, Haryana, India

^{*2}Associate Professor, School of Engineering (SET), Sushant University, Gurugram, Haryana
nehagupta@sushantuniversity.edu.in

ABSTRACT

Forgery of academic and professional certificates remains a major concern across institutions. Traditional centralized systems are prone to manipulation and single points of failure. This work presents a blockchain-based certificate issuance and verification platform developed using Spring Boot and the Ethereum Sepolia test network. The system supports multiple organizations where issuers register and are approved by an administrator before generating certificates. Each certificate is assigned a unique identifier, and a SHA-256 hash of its data is stored on the blockchain through smart contracts. The platform also automates PDF certificate creation with embedded QR codes and sends them via email. Additional features include bulk certificate generation, revocation support, and public verification without requiring a blockchain wallet. Experimental observations indicate an average issuance time of around 4 seconds and verification within 1.5 seconds.

Keywords: Blockchain, Ethereum, Smart Contracts, SHA-256, Certificate Verification, Spring Boot, Web3j, PDF Automation

I. INTRODUCTION

Authenticity of academic credentials plays a crucial role in building trust in education and employment systems. However, fake certificates and manipulated records are increasingly common. Traditional systems, whether paper-based or digital, depend on centralized authorities, making them vulnerable to tampering, data breaches, and system failures. The growing adoption of digital credentials across institutions worldwide has made reliable and scalable verification infrastructure an urgent necessity.

Blockchain technology offers a decentralized and immutable solution where once data is recorded, it cannot be altered. Initially introduced by Nakamoto as the foundation of Bitcoin, blockchain gained further capabilities with Ethereum, enabling programmable smart contracts. The concept of smart contracts as self-executing digital agreements was first formally described by Szabo, and their integration with blockchain has since enabled automated, trustless processing of transactions and credential records. These smart contracts can securely store certificate hashes computed using the SHA-256 standard, ensuring integrity and authenticity.

Despite several proposed solutions, many existing systems lack practical features such as multi-organization support, automated workflows, and user-friendly verification methods. Studies from diverse contexts, including certificate verification challenges in developing regions, accreditation systems, institutional platforms, blockchain-based validation frameworks, and decentralized certification applications, all confirm that no single existing system provides a complete end-to-end solution.

II. LITERATURE REVIEW

A. Gangwar and Chaurasia [4]

Gangwar and Chaurasia [4] developed a blockchain-based certificate authentication system using QR codes and decentralized applications on Ethereum. Their approach improves transparency and enables verification through mobile devices. However, it is limited to a single issuing entity and does not include features such as automated certificate generation, email delivery, or revocation.

B. Sam Prince Franklin et al. [5]

Sam Prince Franklin et al. [5] proposed a decentralized certificate issuance and verification system built on Ethereum, targeting the education, healthcare, and government sectors. While the system effectively demonstrates blockchain-based verification with a mobile interface, it lacks automation features such as PDF generation, multi-issuer support, email delivery, and bulk processing capabilities.

C. Rahman et al. [6]

Rahman et al. [6] presented Verifi-Chain, which integrates blockchain with IPFS to provide decentralized credential verification. Although it improves storage efficiency and security, it follows an applicant-driven model and requires wallet interaction, limiting usability. It also lacks automation and multi-organization features.

D. Certiblock [7]

The authors of Certiblock [7] proposed a prototype that uses blockchain and cloud storage to authenticate academic credentials. Certificates are temporarily held on a server before being transferred to cloud storage, where a unique hash is generated and anchored on the blockchain for verification. However, the system depends on centralized cloud storage rather than a fully decentralized approach and does not support multi-organization issuer management, automated PDF generation, email delivery, bulk issuance, or certificate revocation.

E. IEEE 2025 [8]

This IEEE 2025 paper [8] introduces a role-based blockchain system using Ethereum and IPFS for academic certificate verification. Only the content hash is recorded on-chain while certificate data is stored on IPFS for efficiency. While it demonstrates improved storage management, the system depends on blockchain wallets and does not provide automated certificate generation, email-based delivery, multi-organization issuer management, bulk issuance, or revocation support.

F. Said et al. [10]

Said et al. [10] examined blockchain-based credential verification challenges in developing regions, specifically Tanzania. Their conceptual model proposes a framework to address institutional limitations such as lack of digital infrastructure and absence of standardized verification procedures. While their work identifies key structural barriers and provides a theoretical foundation for blockchain adoption in credential management, it remains at the conceptual stage and does not offer a working implementation, automation capabilities, or multi-organization support.

G. Tariq et al. [11]

Tariq et al. [11] proposed Cerberus, a blockchain-based accreditation and degree verification system targeting higher education institutions. The system uses a permissioned blockchain to provide role-based access to credential data and supports verification by employers and institutions. However, Cerberus requires participants to operate blockchain nodes, increasing infrastructure overhead. It does not support automated PDF issuance, bulk processing, email delivery, or wallet-free public verification, limiting its accessibility for end users.

H. Stefan-Robert and Butincu [12]

Stefan-Robert and Butincu [12] developed a decentralized blockchain-based platform for managing and issuing academic certificates, presented at ICSTCC 2024. Their system emphasizes decentralization and tamper-proof record keeping through smart contracts. While technically sound, the platform does not address multi-organization issuer workflows, admin-based approval mechanisms, automated certificate generation, email delivery, or bulk issuance, making it unsuitable for large-scale institutional deployments without significant additional development.

I. Gayathiri et al. [13]

Gayathiri et al. [13] presented an early blockchain-based certificate validation system that demonstrates the feasibility of recording certificate hashes on a distributed ledger for tamper detection. Although it established foundational proof-of-concept for the approach, the system operates in a single-organization context, lacks automation, and does not provide email delivery, bulk processing, revocation mechanisms, or a public wallet-free verification interface. It serves as an important baseline but does not meet the requirements of modern multi-institution deployments.

J. Jha et al. [14]

Jha et al. [14] proposed Certifier DApp, a decentralized and secured certification system built on blockchain technology, presented at ICBDS 2023. The system uses a decentralized application (DApp) model to issue and verify certificates without a central authority. While the approach demonstrates the viability of DApp-based credentialing, it relies on user-held wallets for interaction, does not support institutional multi-organization

workflows, lacks automated PDF generation or email delivery, and does not provide bulk certificate issuance capabilities.

K. Dewan et al. [16]

Dewan, Duhan, and Gupta [16] proposed a secure electronic voting system using mobile applications and blockchain technology. Their work demonstrates the effectiveness of blockchain in building tamper-proof, transparent, and decentralized systems for sensitive institutional processes. The use of cryptographic verification and mobile-based access in their voting system shares conceptual similarities with the certificate verification approach adopted in this work, reinforcing the applicability of blockchain for securing high-stakes digital records.

L. Dargan, Gupta and Singh [17]

Dargan, Gupta, and Singh [17] proposed a blockchain-based energy management system presented at ICTAI 2021. Their model demonstrates how blockchain can be leveraged to bring transparency, decentralization, and immutability to domain-specific management systems. This cross-domain application of blockchain further validates the core principles employed in the proposed certificate platform, where similar properties — immutability, auditability, and decentralized trust — are applied to academic credential management.

M. Dargan and Gupta [18]

Dargan and Gupta [18] authored a book chapter on the integration of blockchain for IoT communication, published in *Adaptive Power Quality for Power Management using Smart Technologies* (CRC Press, Taylor and Francis, 2023). Their work examines how blockchain protocols can secure communication channels in IoT ecosystems, ensuring data integrity and preventing unauthorized modifications. The secure hash-based communication model discussed in their work aligns with the SHA-256 hash anchoring approach used in this certificate platform for tamper-proof credential storage.

N. Gupta et al. [19][20]

In two related contributions published in *E-Mobility in Electrical Energy Systems for Sustainability* (IGI Global, 2024), the research team examined blockchain's role in emerging smart infrastructure systems [19][20]. These works highlight the broad applicability of blockchain beyond financial transactions — demonstrating its utility in securing critical data across transportation, energy, and communication systems. This interdisciplinary perspective reinforces the motivation for adopting blockchain in the proposed certificate management platform, where trust, transparency, and tamper-resistance are equally essential requirements.

O. Research Gap

Analysis of existing systems reveals six common limitations: (i) lack of multi-organization issuer management with admin approval; (ii) absence of automated certificate generation; (iii) no integrated email delivery system; (iv) missing bulk issuance functionality; (v) dependence on blockchain wallets or external infrastructure; and (vi) limited or absent revocation mechanisms. The proposed system addresses all these limitations within a unified platform.

III. PROPOSED SYSTEM AND METHODOLOGY

A. System Overview

The proposed platform is built using Spring Boot with MySQL for storage and Ethereum Sepolia for blockchain interaction via Web3j. Additional tools include iText for PDF generation, JavaMail for email delivery, and Apache POI for handling bulk uploads. Role-based access control is implemented using Spring Security. The overall system architecture is illustrated in Figure 1.

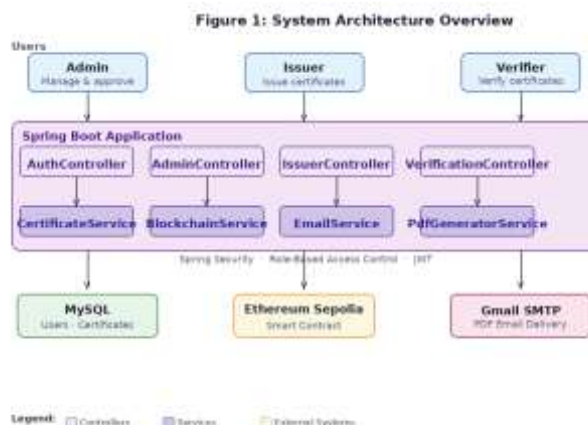


Figure 1: System Architecture Overview

B. User Roles

Administrator: Approves or rejects issuer registrations, monitors all certificates across organizations, and manages revocations.

Issuer: Represents an institution. Registers via a public form and gains access only after admin approval. Generates and manages certificates after approval.

Verifier: Public user who can verify any certificate without creating an account or using a blockchain wallet.

C. Certificate Issuance Process

The certificate issuance process follows a six-stage pipeline as illustrated in Figure 2.

Input Collection: Issuer submits recipient name, email, course title, enrollment number, and passing year.

Duplicate Check: System confirms no prior certificate exists for the same enrollment number from the same issuer.

UUID and SHA-256: A UUID is assigned as certificate ID. SHA-256 is computed over concatenated name, enrollment number, course, and year.

Blockchain Registration: Blockchain Service signs and broadcasts a raw Ethereum transaction calling registerCertificate(string, string) on the deployed Solidity smart contract on the Sepolia testnet.

PDF Generation: PdfGeneratorService renders an A4-landscape PDF with recipient details and a ZXing QR code encoding the public verification URL.

Email Delivery: The PDF is attached to an email sent via Gmail SMTP to the recipient.



Figure 2: Certificate Issuance Pipeline

D. Verification and Bulk Processing

Verification involves checking both database records and blockchain hash consistency, as shown in Figure 3. The public endpoint performs a dual-layer check: database status (ACTIVE or REVOKED) and on-chain confirmation

via eth_call. Bulk issuance is supported through Excel uploads via Apache POI, where each entry is processed independently through the full issuance pipeline.

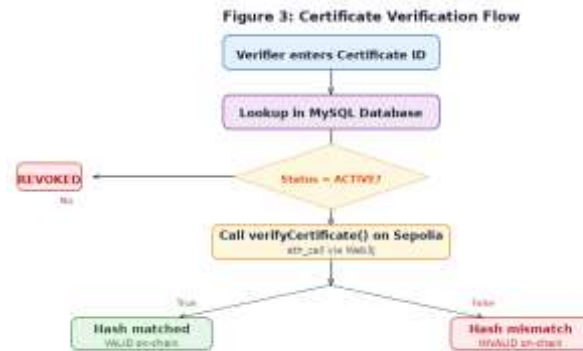


Figure 3: Certificate Verification Flow

IV. COMPARATIVE ANALYSIS

Table I presents a comparative analysis of the proposed system against ten related systems, evaluating six key functional capabilities identified from the literature review.

TABLE I COMPARISON OF RELATED SYSTEMS

System / Feature	Multi-Org	Auto PDF	Email	Bulk Issue	Revoke	Wallet-Free Verify
Gangwar & Chaurasia [4]	No	No	No	No	No	No
Sam Prince Franklin [5]	No	No	No	No	No	No
Verifi-Chain [6]	No	No	No	No	No	No
Certiblock [7]	No	No	No	No	No	No
IEEE 2025 [8]	No	No	No	No	No	No
Said et al. [10]	No	No	No	No	No	No
Cerberus [11]	No	No	No	No	No	No
Stefan-Robert [12]	No	No	No	No	No	No
Gayathiri [13]	No	No	No	No	No	No
Certifier DApp [14]	No	No	No	No	No	No
Proposed System	Yes	Yes	Yes	Yes	Yes	Yes

V. IMPLEMENTATION AND RESULTS

A. Implementation Environment

The system was tested on a local environment running Ubuntu 22.04 LTS with Java 17, Spring Boot, and MySQL. Blockchain transactions were executed on the Ethereum Sepolia public testnet. All measurements were averaged over ten repeated trials.

B. Performance Metrics

TABLE II SYSTEM PERFORMANCE METRICS

Metric	Value	Notes
Avg. certificate issuance time	~4.2 s	Form submit to email sent
Blockchain tx confirmation	~12-15 s	Sepolia avg. block time
Verification response time	< 1.5 s	DB lookup + eth_call
SHA-256 hash generation	< 5 ms	Java MessageDigest
PDF generation (A4 + QR)	~300 ms	iText landscape
Bulk issuance - 10 certs	~45 s	Sequential with email
Email delivery success rate	100% (10/10)	Gmail SMTP, valid addresses

C. System Interface

Figures 4 through 8 illustrate the key interfaces of the implemented system as observed during testing.



Fig. 4: System Home Page



Fig. 5: Admin Dashboard



Fig. 6: Issuer Dashboard with Certificates



Fig. 7: Bulk Certificate Upload

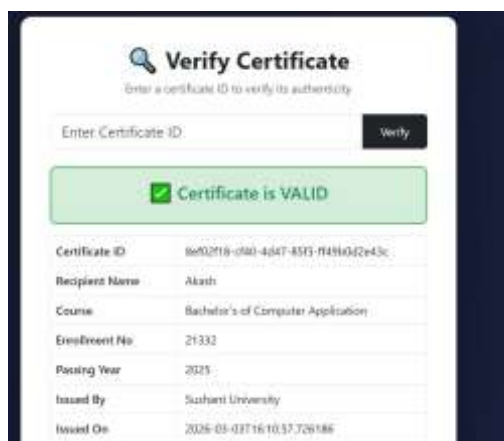


Fig. 8: Certificate Verification Result

D. Security Analysis

Security is ensured through multiple mechanisms. SHA-256 hashing guarantees data integrity — any modification to certificate attributes produces a completely different hash that will not match the on-chain record. Blockchain immutability prevents registered hashes from being deleted or altered. Role-based access control restricts unapproved issuers and separates admin from issuer functions. Duplicate enrollment checks prevent double-issuance. The revocation mechanism provides database-level governance without modifying the immutable on-chain record. These security properties are consistent with the access control model proposed by Tariq et al. [11] in Cerberus and extend the foundational hash-based tamper detection demonstrated by Gayathiri et al. [13], while adding multi-organization governance and automated workflow capabilities absent in those works.

E. Limitations

The current system uses the Sepolia testnet rather than Ethereum mainnet, meaning it does not incur real transaction costs. Bulk certificate processing is sequential, which may be slow for large datasets. The PDF certificate template currently offers limited customization options for different issuing organizations. These are clear directions for future improvement.

VI. CONCLUSION

This paper presented a complete blockchain-based certificate management system addressing key limitations of existing approaches. By integrating automation, multi-organization support, and user-friendly verification, the system provides a practical and scalable solution. The proposed platform uniquely combines multi-organization issuer management, automated certificate generation, email delivery, bulk processing, revocation, and wallet-free verification within a single unified application.

Comparative analysis against ten related peer-reviewed systems confirmed the platform's unique completeness. Works such as Said et al. [10], Stefan-Robert and Butincu [12], and Jha et al. [14] each address isolated aspects of credential verification but none provide the combined automation, multi-organization governance, and accessibility achieved here. The broad review by Ocheja et al. [15] further validates that the integration of blockchain with institutional workflows remains an underexplored area, which this system directly addresses. Experimental results demonstrate efficient performance with issuance in approximately 4 seconds and verification in under 1.5 seconds. Future improvements include deployment on mainnet or Layer-2 solutions such as Polygon, parallel processing for bulk operations, enhanced certificate customization, IPFS-based file storage, and a mobile-based QR verification application.

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